Hybrid Intelligence for Product Success Prediction: A Survey of Sentiment-Driven Machine Learning in E-Commerce

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***Abstract*—**Forecasting new product success in digital marketplaces is a high-stakes challenge where historical sales signals are unavailable at launch. This survey synthesises 28 peer-reviewed studies published between 2018 and 2025 across IEEE, ACL, Springer, ScienceDirect, and Elsevier to examine the convergence of Natural Language Processing and tabular predictive modelling for e-commerce viability forecasting. Incorporating the PRISMA systematic review by Daza et al. (2024), we examine key methodological patterns: Support Vector Machines and LSTMs dominate historical text mining, while F1-score is the most frequently reported evaluation metric (used in 75% of surveyed studies). However, traditional sentiment analysis evaluates post-launch customer feedback. To address the launch-blind cold-start problem, we survey hybrid architectures that bridge unstructured text and structured regressors. We examine a two-stage hybrid framework: (1) open-weight and hosted LLMs perform null-forcing Aspect-Based Sentiment Analysis (ABSA) on customer reviews to establish category reference distributions, and (2) a skeptic-prompted bridging layer projects candidate product specifications into projected aspect scores, which are fed into a gradient-boosted regressor (XGBoost). Evaluated on 43,599 customer reviews across 631 verified physical products in five categories, this architecture achieves Spearman $\rho = 0.317$ (AUC = 0.743) on a pre-registered holdout and $\rho = 0.545$ (AUC = 0.901) on a temporal holdout. We identify critical research frontiers in specification-to-aspect bridging fidelity, domain adaptation, and calibrated uncertainty quantification.

Index Terms*—Aspect-based sentiment analysis, XGBoost, large language models, e-commerce prediction, launch-blind forecasting, specification bridging, PRISMA systematic review.*

I. INTRODUCTION

Forecasting product success in e-commerce marketplaces is fundamentally a data-driven prediction problem. Digital platforms log customer opinions, purchase histories, pricing trajectories, and engagement signals at an unprecedented scale [1]. However, a fundamental temporal asymmetry exists: while post-launch products accumulate rich behavioral telemetry, prospective products require capital commitments long before any consumer feedback exists. A product manager or direct-to-consumer (D2C) founder launching a new stock keeping unit (SKU) must evaluate viability at the 'launch-blind' stage—where only specifications, target pricing, and category norms are observable [2].

Recent advances in machine learning provide two complementary capabilities to address this challenge. First, gradient boosted decision trees, particularly XGBoost [3], have established strong empirical baselines for tabular e-commerce prediction tasks. Second, Large Language Models (LLMs) [4]–[6] have demonstrated remarkable zero-shot aspect extraction and semantic reasoning capabilities. A recent PRISMA-based systematic literature review by Daza et al. [1] surveyed 20 e-commerce sentiment studies (2018–2024), confirming that while Support Vector Machines (SVM) and Long Short-Term Memory (LSTM) networks have historically dominated the literature, F1-score is the most frequently reported evaluation metric (used in 75% of studies). Nevertheless, traditional pipelines remain post-launch evaluators.

Within the 28 peer-reviewed studies identified in this survey, this paper synthesizes the methodological landscape bridging unstructured natural language processing and structured tabular regressors. We examine a two-stage hybrid architecture: Stage 1 extracts aspect sentiment from 43,599 historical customer reviews across 631 verified physical products (quarantining fraudulent reviews) to construct category reference priors; Stage 2 maps candidate pre-launch specifications into aspect space via skeptic prompting, predicting eventual market success via a monotonic-constrained gradient boosted regressor.

II. LITERATURE REVIEW: SENTIMENT ANALYSIS AS A PREDICTIVE SIGNAL

A. Classical and Lexicon-Based Approaches

The foundations of opinion mining were established by Pang and Lee [7], who categorized sentiment analysis into document, sentence, and aspect levels. Early e-commerce systems relied on curated lexical databases such as SentiWordNet [8]. Hu and Liu [9] introduced aspect extraction from customer reviews, showing that aggregate star ratings discard critical multi-dimensional quality indicators. Daza et al. [1] formalize this taxonomy: Aspect-Level Sentiment Analysis (ASA) identifies specific product entities and their associated polarities. One illustrative formulation of lexicon scoring is:

$$\text{Sentiment}(d) = (1/n)\, \Sigma_i \, [\, \text{Polarity}(w_i) \times \text{Weight}(w_i) \,] \qquad (1)$$

where d is a review text, w_i represents constituent tokens, and $\text{Weight}(w_i)$ denotes the term significance. While computationally lightweight, lexicon approaches suffer from semantic fragility in colloquial product feedback.

B. Transformer Models and Transfer Learning

The introduction of BERT [4] shifted sentiment analysis toward contextualized representation learning. Shan et al. [10] benchmarked five model families on bilingual e-commerce reviews, showing that ELECTRA achieved 98.09% classification accuracy (F1 = 0.9711), outperforming standard BERT (95.28%) and BiLSTM (96.09%). Wang et al. [11] developed Multi-Feature Fusion Graph Attention Networks (MFF-GAT) for aspect sentiment classification, effectively integrating syntactic dependency trees with semantic attention. In financial e-commerce, Sun [12] utilized FinBERT on social commentary, demonstrating that sentiment polarity correlates positively with transaction volume.

III. GRADIENT BOOSTING AND ENSEMBLE METHODS FOR E-COMMERCE PREDICTION

Tabular data in retail e-commerce exhibits complex non-linear relationships, mixed categorical-continuous distributions, and heterogeneous feature scales. Chen and Guestrin's XGBoost [3] employs regularized gradient boosting with second-order Taylor expansions of the loss function:

$$\text{Obj}(\theta) = \Sigma_i \, L(y_i, \hat{y}_i) + \Sigma_t \, [\, \gamma \, T_t + (1/2)\lambda \, \|w_t\|^2 \,] \qquad (2)$$

Sayyad et al. [13] demonstrated the efficacy of CatBoost for supply chain optimization under extreme categorical sparsity. Shang [14] integrated CART decision trees with high-dimensional sentiment indicators to forecast price volatility. In the systematic review by Daza et al. [1], XGBoost appeared in only 10% of studies despite achieving top accuracy (91.60% in Elangovan and Subedha [15]), establishing tree-based boosting as a highly effective yet underutilized backbone.

IV. HYBRID ARCHITECTURES: BRIDGING UNSTRUCTURED TEXT TO REGRESSORS

Emerging architectures increasingly connect natural language feature extractors to tabular estimators. Hidayat et al. [16] demonstrated Large Language Model integration for time-series discrepancy prediction, analyzing probabilistic versus point forecasting tradeoffs. Niimi [17] investigated local LLM sentiment analysis with majority voting across temperature variations, finding that multi-pass inference improves classification consistency. However, rather than stochastic majority voting, production pipelines achieve deterministic reliability through constrained JSON schema enforcement (e.g., Pydantic v2). Under an independent trial model where single-pass JSON validity is $p \approx 0.85$, a three-pass retry policy achieves a theoretical bound of $P(\text{valid}) = 1 - (1 - p)^3 \approx 0.997$, while empirical schema compliance in production exceeds 99.4%.

V. PROPOSED METHODOLOGY: LAUNCH-BLIND SPECIFICATION BRIDGING

A. Two-Stage Learning and Bridging Framework

Traditional e-commerce models assume reviews are available. LaunchSight resolves the cold-start dilemma through a two-stage learning and bridging framework (Fig. 1). Stage 1 constructs the empirical foundation: historical customer reviews across five physical product categories are processed using an LLM with null-forcing Aspect-Based Sentiment Analysis. Aspect scores are extracted across six shared dimensions (Value for Money, Utility, Ease of Use, Reliability, Design Appeal, After-Sales) and three physical dimensions (Build Quality, Durability, Repairability). This establishes empirical category priors.

Stage 2 performs launch-blind specification bridging. When evaluating a prospective product before launch, no customer reviews exist. A skeptic-prompted LLM evaluates the manufacturer's technical specifications, pricing, and category context, mapping unstructured specifications into estimated aspect scores grounded against the category reference distribution. These bridged aspects feed a trained XGBoost regressor enforcing monotonic constraints, outputting a composite viability prediction.

B. Dataset Provenance and Fraud Filtering

The empirical corpus comprises 43,599 verified consumer reviews spanning 640 physical products across five Amazon categories: Bluetooth speakers, ice makers, power banks, smartwatches, and wireless headphones. Mobile applications, examined during preliminary exploratory feasibility, were excluded from the core production methodology due to platform policy divergences and distinct telemetry. To safeguard against manipulation, an automated review fraud filter removed 9 suspect products exhibiting sudden review volume bursts or extreme J-shaped rating anomalies, leaving a clean training corpus of 631 physical products.

C. Ground Truth Composite Success Metric

To establish a defensible target variable without succumbing to single-star rating biases, market success is formulated as a three-component composite score normalized within each category:

$$\text{SuccessScore} = 100 \times [\ 0.40 \times \text{Volume_norm} + 0.35 \times \text{Rating_norm} + 0.25 \times \text{Velocity_norm}\ ] \quad (3)$$

where Volume denotes log1p(review_count), Rating reflects average star rating, and Velocity captures reviews accumulated per month. All scaling parameters are computed strictly on training folds to prevent target leakage.

D. Boundary Conditions and the Ice Maker Paradox

A rigorous prediction methodology must explicitly delineate its boundary conditions. The LaunchSight evaluation uncovered the 'Ice Maker Paradox': while ice makers exhibited strong review-to-success correlation (mean $|r| = 0.32$), specs-only bridging achieved near-zero rank correlation ($\rho \approx 0.04$). Diagnostic analysis revealed that ice maker specifications consist almost exclusively of physical capacity and cycle metrics (e.g., '26 lb/day, 9 cubes in 6 minutes'), which carry zero information regarding compressor longevity, pump durability, or operational noise. In contrast, personal electronics specifications (headphones, smartwatches) encode rich experiential attributes (battery chemistry, active noise cancellation, driver materials), yielding high bridging fidelity ($\rho = 0.50$–0.53). LaunchSight implements calibrated runtime abstention when specification expressiveness falls below validated thresholds.

VI. COMPARATIVE MODEL PERFORMANCE SUMMARY

To avoid metric incompatibility (e.g., contrasting sentence-level sentiment accuracy against product-level viability AUC), Table I summarizes the empirical landscape across two distinct tasks: (1) NLP sentiment classification reported in recent e-commerce literature, and (2) Launch-Blind Product Viability Prediction benchmarked in this research.

Study / Architecture	Task Domain	Primary Technique	Evaluation Metric	Reported Result
Shan et al. [10] (2025)	Bilingual E-Commerce Review Polarity	ELECTRA Transformer	Accuracy, F1-Score	Acc = 98.09%, F1 = 0.9711
Thomas & Jeba [18] (2024)	Multiclass Review Sentiment	LSTM + Collab. Filtering	Accuracy, F1-Score	Acc = 98.43%, F1 = 0.9820
Bin Harunasir et al. [19] (2023)	Amazon Product Review Classification	Supervised LSTM	Accuracy, Precision	Acc = 97.00%, Prec = 96.50%
Elangovan & Subedha [15] (2023)	Large-Scale Amazon Sentiment (100k)	XGBoost Classifier	Accuracy, MCC	Acc = 91.60%, MCC = 0.842
Wang et al. [11] (2025)	Fine-Grained ABSA Aspect Classification	MFF-GAT (Graph Attn)	Macro F1-Score	Outperformed 5 ABSA baselines
LaunchSight (In-Sample	Aspect-to-Success Viability	XGBoost Regressor	5-Fold CV R^2, RMSE, Acc	$R^2 = 0.566$, RMSE = 7.1,

CV)	Modeling	(n=631)		Acc = 81.0%
LaunchSight (Frozen Holdout)	Launch-Blind Viability from Specs	Spec Bridging + XGBoost (n=59)	Spearman ρ, Top/Bottom AUC	ρ = 0.317 (p=0.014), AUC = 0.743
LaunchSight (Temporal Holdout)	Forward-in-Time Viability (>Oct 2021)	Spec Bridging + XGBoost (n=66)	Spearman ρ, Top/Bottom AUC	ρ = 0.545, AUC = 0.901

TABLE I. COMPARATIVE PERFORMANCE: LITERATURE SENTIMENT CLASSIFICATION VS. LAUNCH-BLIND VIABILITY RANKING.

VII. SYNTHESIS AND OPEN RESEARCH FRONTIERS

Four central insights emerge from this synthesis: (1) Sentiment analysis in e-commerce has transitioned from lexicon matching to transformer-based aspect extraction, enabling nuanced multi-dimensional feature representations. (2) Gradient boosted decision trees (XGBoost, CatBoost) remain the most effective modeling frameworks for tabular retail prediction, offering computational efficiency, robustness to outliers, and native monotonic constraint support. (3) Within the studies surveyed, prior research focused almost exclusively on retrospective post-launch reviews; Launch-Blind specification bridging represents a critical methodology for pre-manufacturing evaluation. (4) Bridging fidelity is fundamentally bounded by specification expressiveness, demanding automated abstention mechanisms when input copy lacks experiential signal.

VIII. CONCLUSION

This survey synthesized 28 peer-reviewed studies across natural language processing and e-commerce machine learning. By integrating the bibliometric findings of Daza et al. [1] with the LaunchSight two-stage hybrid framework, we demonstrated how aspect-level sentiment extraction from historical customer reviews can establish robust category reference priors, enabling launch-blind viability forecasting from technical product specifications alone. Validated across 43,599 customer reviews and 631 verified physical products, this approach bridges the critical cold-start gap in modern retail analytics.

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